



TM CIP RB (AFCO 5665)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations
Revision Date: 01/27/2025

Version: 1.0

SECTION 1: IDENTIFICATION

Product Identifier

Product Form: Mixture

Product Name: TM CIP RB

Product Code: AFCO 5665

Intended Use of the Product

Use of the Substance/Mixture: Alkaline CIP and membrane cleaner for the food industry. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company

AFCO

550 Development Avenue

Chambersburg, PA 17201

T: 800-345-1329

www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

Met. Corr. 1 H290

Skin Corr. 1A H314

Eye Dam. 1 H318

Aquatic Acute 3 H402

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US)

:



GHS05

Signal Word (GHS-US)

: Danger.

Hazard Statements (GHS-US)

: H290 - May be corrosive to metals
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H402 - Harmful to aquatic life

Precautionary Statements (GHS-US)

: P234 - Keep only in original container.
P260 - Do not breathe vapors, mist, spray.
P264 - Wash hands, forearms, and other exposed body parts thoroughly after handling.
P280 - Wear protective glove, protective clothing, eye protection, face protection.
P301+P330+P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P303+P361+P353 - IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338 - If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 - Immediately call a POISON CENTER or doctor/physician
P321 - Specific treatment (see section 4).
P363 - Wash contaminated clothing before reuse.
P390 - Absorb spillage to prevent material damage.
P405 - Store locked up.

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P406 - Store in corrosive resistant container with a resistant inner liner.

P501 - Dispose of contents/container according to local, regional, national, territorial, provincial, and international regulations.

Other Hazards

Results of PBT and vPvB Assessment: This mixture does not contain any substances that are assessed to be a PBT or a vPvB.

Unknown Acute Toxicity (GHS-US): May be harmful if swallowed (GHS category 5: acutely toxic - oral).

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances: Not relevant (mixture).

Mixture

Name	Product identifier	% (w/w)	Classification (GHS-US)
Water	(CAS No) 7732-18-5	55-79	Not classified.
Potassium hydroxide	(CAS No) 1310-58-3	5 - 10	Met. Corr. 1, H290 Acute Tox. 4 (Oral), H302 Skin Corr. 1A, H314 Eye Dam. 1, H318
Sodium hydroxide	(CAS No) 1310-73-2	10-20	Met. Corr. 1, H290 Skin Corr. 1A, H314 Eye Dam. 1, H318 Aquatic Acute 3, H402
Tetrasodium EDTA	(CAS No) 64-02-8	5-10	Acute Tox. 4 (Oral), H302 Acute Tox. 4 (Inhalation: dust, mist), H332 Eye Dam. 1, H318 Aquatic Acute 2, H401
D-Glucopyranose, oligomeric, decyl octyl glycosides	(CAS No) 68515-73-1	1-5	Eye Dam. 1, H318

Full text of H-phrases: see section 16.

SECTION 4: FIRST AID MEASURES

Description of First Aid Measures

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

Inhalation: When symptoms occur: go into open air and ventilate suspected area. Remove to fresh air and keep at rest in a position comfortable for breathing. Obtain medical attention if breathing difficulty persists.

Skin Contact: Remove contaminated clothing. Drench affected area with water for at least 15 minutes. Obtain medical attention if irritation persists. Wash contaminated clothing before reuse.

Eye Contact: Rinse cautiously with water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Obtain medical attention if irritation develops or persists.

Ingestion: Rinse mouth. Do NOT induce vomiting. Immediately call a POISON CENTER or doctor/physician.

Most Important Symptoms and Effects Both Acute and Delayed

Symptoms and effects are not known to date.

Indication of Any Immediate Medical Attention and Special Treatment Needed

None.

SECTION 5: FIRE-FIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Water spray, carbon dioxide, alcohol resistant foam, dry chemical.

Unsuitable Extinguishing Media: Do not use a heavy water stream. Use of heavy stream of water may spread fire.

Special Hazards Arising From the Substance or Mixture

Fire Hazard: Not flammable.

Explosion Hazard: Product is not explosive.

Reactivity: When heated to decomposition, emits toxic fumes. Corrosive vapors are released. Substance or mixture is corrosive to soft metals.

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Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire.

Firefighting Instructions: Use water spray or fog for cooling exposed containers.

Protection During Firefighting: Do not enter fire area without proper protective equipment, including respiratory protection.

Hazardous Combustion Products: Carbon oxides (CO, CO₂). Nitrogen oxides (NO_x). Sodium oxides. Potassium oxides.

Other information: Do not allow run-off from fire fighting to enter drains or water courses.

Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Avoid breathing (vapors, mist, spray). Avoid contact with skin, eyes, or clothing.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate unnecessary personnel.

For Emergency Personnel

Protective Equipment: Equip cleanup crew with proper protection.

Emergency Procedures: Ventilate area.

Environmental Precautions

Prevent entry to sewers and public waters. Avoid release to the environment.

Methods and Material for Containment and Cleaning Up

For Containment: Contain any spills with dikes or absorbents to prevent migration and entry into sewers or streams.

Methods for Cleaning Up: Clear up spills immediately and dispose of waste safely. Contact competent authorities after a spill. Wipe up with absorbent material (e.g. cloth, fleece). Collect spillage: sawdust, kieselgur (diatomite), sand, universal binder, absorbents, neutralizing agents.

Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Additional Hazards When Processed: When heated to decomposition, emits toxic fumes. Corrosive vapors are released.

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do not eat, drink or smoke when using this product. Wash hands and forearms thoroughly after handling.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Comply with applicable regulations.

Storage Conditions: Store in a dry, cool and well-ventilated place. Keep container closed when not in use. Keep/Store away from extremely high or low temperatures, incompatible materials.

Incompatible Materials: Strong acids. Soft metals.

Recommended Storage Temperature: 5 – 25°C/41 – 77°F.

Specific End Use(s): Alkaline CIP and membrane cleaner for the food industry. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Potassium hydroxide (1310-58-3)		
USA ACGIH	ACGIH Ceiling (mg/m ³)	2 mg/m ³
USA NIOSH	NIOSH REL (ceiling) (mg/m ³)	2 mg/m ³
Ontario	OEL Ceiling (mg/m ³)	2 mg/m ³
Québec	PLAFOND (mg/m ³)	2 mg/m ³
Sodium hydroxide (1310-73-2)		
Mexico	OEL Ceiling (mg/m ³)	2 mg/m ³
USA ACGIH	ACGIH Ceiling (mg/m ³)	2 mg/m ³
USA OSHA	OSHA PEL (TWA) (mg/m ³)	2 mg/m ³
USA NIOSH	NIOSH REL (ceiling) (mg/m ³)	2 mg/m ³
USA IDLH	US IDLH (mg/m ³)	10 mg/m ³

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Ontario	OEL Ceiling (mg/m ³)	2 mg/m ³
Québec	Plafond (mg/m ³)	2 mg/m ³

Exposure Controls

Appropriate Engineering Controls: Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Ensure all national/local regulations are observed. Ensure adequate ventilation, especially in confined areas.

Personal Protective Equipment: Protective goggles. Protective clothing. Gloves. Use protective eyewear to guard against splash of liquids.



Materials for Protective Clothing: Chemically resistant materials and fabrics.

Hand Protection: Wear chemically resistant protective gloves.

Eye Protection: Chemical goggles or face shield.

Skin and Body Protection: Wear suitable protective clothing.

Respiratory Protection: A respirator is not needed under normal and intended conditions of use. In case of inadequate ventilation wear respiratory protection.

Other Information: When using, do not eat, drink or smoke.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	: Liquid.
Appearance	: Various.
Odor	: Characteristic.
Odor Threshold	: Not determined.
pH	: 12.4 – 13 (1%) (base)
Relative Evaporation Rate (butylacetate=1)	: Not determined.
Melting Point	: Not determined.
Freezing Point	: Not determined.
Boiling Point	: 100°C (212°F)
Flash Point	: None to boiling.
Auto-ignition Temperature	: Not determined.
Decomposition Temperature	: Not determined.
Flammability (solid, gas)	: Not relevant.
Lower Flammable Limit	: Not determined
Upper Flammable Limit	: Not determined.
Vapor Pressure	: 32 hPa at 25°C
Relative Vapor Density at 20°C	: Not determined.
Specific Gravity	: 1.35 – 1.37
Solubility	: Soluble in water.
Partition coefficient: n-octanol/water	: Not determined
Viscosity	: Not determined.
Explosion Data	: None.
Oxidizing Data	: None.

SECTION 10: STABILITY AND REACTIVITY

Reactivity: Contact with soft metals may evolve flammable hydrogen gas.

Chemical Stability: Stable at standard temperature and pressure.

Possibility of Hazardous Reactions: Hazardous polymerization will not occur.

Conditions to Avoid: Incompatible materials.

Incompatible Materials: Strong acids. Soft metals.

Hazardous Decomposition Products: Carbon oxides (CO, CO₂). Nitrogen oxides (NO_x). Sodium oxides. Potassium oxides.

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SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects - Product

Acute Toxicity: Not classified.

LD50 and LC50 Data: Not available.

Skin Corrosion/Irritation: Causes severe skin burns and eye damage.

Serious Eye Damage/Irritation: Causes serious eye damage.

Respiratory or Skin Sensitization: Not classified.

Germ Cell Mutagenicity: Not classified.

Teratogenicity: Not classified.

Carcinogenicity: Not classified.

Specific Target Organ Toxicity (Repeated Exposure): Not classified.

Reproductive Toxicity: Not classified.

Specific Target Organ Toxicity (Single Exposure): Not classified.

Aspiration Hazard: Not classified.

Information on Toxicological Effects - Ingredient(s)

LD50 and LC50 Data:

Water (7732-18-5)	
LD50 Oral Rat	>90000 mg/kg
Potassium hydroxide (1310-58-3)	
LD50 Oral Rat	214 mg/kg
ATE (oral)	333.000 mg/kg body weight
Tetrasodium EDTA (64-02-8)	
LD50 Oral Rat	1780 mg/kg
ATE US (dust, mist)	1.50 mg/l/4h
D-Glucopyranose, oligomeric, decyl octyl glycosides (68515-73-1)	
LD50 Oral Rat	>5000 mg/kg

SECTION 12: ECOLOGICAL INFORMATION

Toxicity

General: Harmful to aquatic life.

Sodium hydroxide (1310-73-2)	
LC50 Fish 1	40 mg/l
Tetrasodium EDTA (64-02-8)	
LC50 Fish 1	41 mg/l (Exposure time: 96 h - Species: Lepomis macrochirus [static])
LC50 Fish 2	59.8 mg/l (Exposure time: 96 h - Species: Pimephales promelas [static])
ErC50 (algae)	2.77 mg/l (72 h - Species: Desmodesmus subspicatus)
EC50 Daphnia 2	0.033 (0.033 - 0.044) mg/l (Exposure time: 48 h - Species: Daphnia magna [Static])
D-Glucopyranose, oligomeric, decyl octyl glycosides (68515-73-1)	
LC50 Fish 1	6-198 mg/l (Exposure time: 96 h - Species: Brachydanio rerio)
EC50 Daphnia 1	14-294 mg/l (48 h - Species: Daphnia magna)

Persistence and Degradability

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Persistence and Degradability	Not established.

Bioaccumulative Potential

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Bioaccumulative Potential	Not established.
Tetrasodium EDTA (64-02-8)	
Log Pow	5.01 (calculated)
Potassium hydroxide (1310-58-3)	
Log Pow	0.65

Mobility in Soil Not available.

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Other Adverse Effects

Other Information: Not available.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Recommendations: Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations.

SECTION 14: TRANSPORT INFORMATION

14.1 In Accordance with DOT

Proper Shipping Name : CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S. (Potassium hydroxide, sodium hydroxide)
Hazard Class : 8
Identification Number : UN3266
Label Codes : 8
Packing Group : II
ERG Number : 154



14.2 In Accordance with IMDG

Proper Shipping Name : CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S. (Potassium hydroxide, sodium hydroxide)
Hazard Class : 8
Identification Number : UN3266
Packing Group : II
Label Codes : 8
EmS-No. (Fire) : F-A
EmS-No. (Spillage) : S-B



14.3 In Accordance with IATA

Proper Shipping Name : CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S. (Potassium hydroxide, sodium hydroxide)
Packing Group : II
Identification Number : UN3266
Hazard Class : 8
Label Codes : 8
ERG Code (IATA) : 8L



14.4 In Accordance with TDG

Proper Shipping Name : CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S. (Potassium hydroxide, sodium hydroxide)
Packing Group : II
Hazard Class : 8
Identification Number : UN3266
Label Codes : 8



SECTION 15: REGULATORY INFORMATION

US Federal Regulations

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SARA Section 311/312 Hazard Classes	Immediate (acute) health hazard.
Water (7732-18-5)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Potassium hydroxide (1310-58-3)	
Listed on the United States TSCA (Toxic Substances Control Act) inventor.	
Tetrasodium EDTA (64-02-8)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
D-Glucopyranose, oligomeric, decyl octyl glycosides (68515-73-1)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Sodium hydroxide (1310-73-2)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	

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US State Regulations

Sodium hydroxide (1310-73-2)
U.S. - California - SCAQMD - Toxic Air Contaminants - Non-Cancer Acute. U.S. - California - Toxic Air Contaminant List (AB 1807, AB 2728). U.S. - New Jersey - Discharge Prevention - List of Hazardous Substances. RTK - U.S. - New Jersey - Right to Know Hazardous Substance List. U.S. - New Jersey - Special Health Hazards Substances List. U.S. - New York - Occupational Exposure Limits – TWAs. U.S. - New York - Reporting of Releases Part 597 - List of Hazardous Substances. RTK - U.S. - Pennsylvania - RTK (Right to Know) - Environmental Hazard List. RTK - U.S. - Pennsylvania - RTK (Right to Know) List. U.S. - Texas - Effects Screening Levels - Long Term. U.S. - Texas - Effects Screening Levels - Short Term.
Potassium hydroxide (1310-58-3)
U.S. - New Jersey - Discharge Prevention - List of Hazardous Substances. U.S. - New Jersey - Right to Know Hazardous Substance List. U.S. - New Jersey - Special Health Hazards Substances List. U.S. - New York - Reporting of Releases Part 597 - List of Hazardous Substances. U.S. - Pennsylvania - RTK (Right to Know) - Environmental Hazard List. U.S. - Pennsylvania - RTK (Right to Know) List. U.S. - Texas - Effects Screening Levels - Long Term. U.S. - Texas - Effects Screening Levels - Short Term.
Tetrasodium EDTA (64-02-8)
U.S. - Texas - Effects Screening Levels - Long Term. U.S. - Texas - Effects Screening Levels - Short Term.
D-Glucopyranose, oligomeric, decyl octyl glycosides (68515-73-1)
U.S. - Texas - Effects Screening Levels - Long Term. U.S. - Texas - Effects Screening Levels - Short Term.

Canadian Regulations

Water (7732-18-5)
Listed on the Canadian DSL (Domestic Substances List) inventory.
Potassium hydroxide (1310-58-3)
Listed on the Canadian DSL (Domestic Substances List) inventory.
Sodium hydroxide (1310-73-2)
Listed on the Canadian DSL (Domestic Substances List) inventory.
Tetrasodium EDTA (64-02-8)
Listed on the Canadian DSL (Domestic Substances List) inventory.
D-Glucopyranose, oligomeric, decyl octyl glycosides (68515-73-1)
Listed on the Canadian DSL (Domestic Substances List) inventory.

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision date	: 01/27/2025
Other Information	: This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

GHS Full Text Phrases:

Acute Tox. 4 (Oral)	Acute toxicity (oral) Category 4.
Acute Tox. 4 (Inhalation)	Acute toxicity (Inhalation; dust, mist) Category 4.
Aquatic Acute 2	Hazardous to the aquatic environment - Acute Hazard Category 2.
Aquatic Acute 3	Hazardous to the aquatic environment - Acute Hazard Category 3.
Eye Dam. 1	Serious eye damage/eye irritation Category 1.
Skin Corr. 1A	Skin corrosion/irritation Category 1A.

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Met. Corr. 1	Corrosive to metals Category 1.
H290	May be corrosive to metals.
H302	Harmful if swallowed.
H314	Causes severe skin burns and eye damage.
H318	Causes serious eye damage.
H332	Harmful if inhaled.
H401	Toxic to aquatic life.
H402	Harmful to aquatic life.

- NFPA Health Hazard** : 3 - Short exposure could cause serious temporary or residual injury even though prompt medical attention was given.
- NFPA Fire Hazard** : 1 - Must be preheated before ignition can occur.
- NFPA Reactivity** : 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.



HMIS III Rating

- Health** : 3 Serious Hazard - Major injury likely unless prompt action is taken and medical treatment is given.
- Flammability** : 1 - Slight Hazard.
- Physical** : 0 - Minimal Hazard.

Party Responsible for the Preparation of This Document

AFCO
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Chambersburg, PA 17201
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This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

North America GHS SDS 2015 (U.S., Can., Mex.)

Disclaimer:

This information is based upon the present state of our knowledge. This SDS has been compiled and is solely intended for this product.